

Pricing American Put Options with Least-Squares Monte Carlo

A Study of Longstaff–Schwartz and Tsitsiklis–Van Roy

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Project Objective

Introduction

American options and Monte Carlo methods

Longstaff–Schwartz (2001) method

- **Key idea:** Use least-squares regression to estimate the continuation value of the option at each exercise date.
- **Steps:**
 1. Simulate multiple paths of the underlying asset price using Monte Carlo methods.
 2. At each exercise date, regress the discounted future payoffs on a set of basis functions of the underlying asset price to estimate the continuation value.
 3. Determine the optimal exercise strategy by comparing the immediate exercise payoff with the estimated continuation value.
- **Advantages:**
 - Can handle high-dimensional problems and complex payoff structures.
 - Provides a flexible framework for pricing American options.
- **Limitations:**
 - The choice of basis functions can significantly affect the accuracy of the estimates.
 - May require a large number of simulations to achieve accurate results,

Tsitsiklis–Van Roy (2001) method

- **Key idea:** Use regression methods to directly approximate the value function of the option, rather than the continuation value.
- **Steps:**
 1. Simulate multiple paths of the underlying asset price using Monte Carlo methods.
 2. At each exercise date, regress the discounted future payoffs on a set of basis functions of the underlying asset price to estimate the value function.
 3. Determine the optimal exercise strategy by comparing the immediate exercise payoff with the estimated value function.
- **Advantages:**
 - Can be more efficient than the Longstaff–Schwartz method in certain cases, especially when the payoff structure is simpler.
 - Provides a direct approximation of the option's value, which can be useful for risk management and hedging purposes.
- **Limitations:**
 - Similar to the Longstaff–Schwartz method, the choice of basis functions can significantly affect the accuracy of the estimation.

Conceptual comparison

Basis functions and regression techniques

- **Basis functions:** Common choices include polynomials, splines, and orthogonal functions. The choice of basis functions can significantly impact the accuracy of the estimates, and may require experimentation to find the best fit for a given problem.
- **Regression techniques:** Both methods typically use least-squares regression to estimate the continuation value or value function. However, alternative regression techniques, such as ridge regression or lasso regression, can be employed to improve the stability and accuracy of the estimates, especially in high-dimensional settings or when the number of basis functions is large.

Basis Functions in Longstaff–Schwartz

- **Common choices:** Polynomials (e.g., linear, quadratic, cubic), splines, and orthogonal functions (e.g., Hermite polynomials).
- **Impact on accuracy:** The choice of basis functions can significantly affect the accuracy of the estimates. For example, using too few basis functions may lead to underfitting, while using too many can lead to overfitting and increased computational complexity. It is often necessary to experiment with different sets of basis functions to find the best fit for a given problem.

Basis Functions in Tsitsiklis–Van Roy

- **Common choices:** Similar to the Longstaff–Schwartz method, common basis functions include polynomials, splines, and orthogonal functions. However, since the Tsitsiklis–Van Roy method directly approximates the value function, the choice of basis functions may need to be more carefully considered to capture the relevant features of the option's value, especially in cases with complex payoff structures or high-dimensional problems.
- **Impact on accuracy:** As with the Longstaff–Schwartz method, the choice of basis functions can significantly affect the accuracy of the estimates. In the Tsitsiklis–Van Roy method, the accuracy of the value function approximation can be particularly sensitive to the choice of basis functions, as it directly impacts the estimation of the option's value and the resulting exercise strategy.

Convergence and accuracy

- **Convergence:** Both methods rely on the law of large numbers to ensure that the estimates converge to the true option value as the number of simulations increases. However, the rate of convergence can be slow, especially for options with long maturities or high volatility, which may require a large number of simulations to achieve accurate results.
- **Accuracy:** The accuracy of the estimates can be affected by several factors, including the choice of basis functions, the number of simulations, and the complexity of the payoff structure. In general, both methods can provide accurate estimates for American options, but may require careful tuning and experimentation to achieve the best results for a given problem.

Experimental Design

- **Objective:** To compare the performance of the Longstaff–Schwartz and Tsitsiklis–Van Roy methods in pricing American vanilla put options, American-Asian-Bermudan options, and American options on assets which follow jump-diffusion processes.

Integrated Implementation Workflow

Benchmark - What the reference methods are and why

Black-Scholes for European options

Implicit finite difference method for American options

Auxiliary method - CRR binomial tree method for American options

Results

Vanilla Benchmark

Exercise Policy Comparison

Robustness Analysis

Asian-Style Options

Jump-Diffusion Options

Longstaff vs. Tsitsiklis: A Direct Comparison

Implementation and Computational Testing

Critical Assessment of Methods

- **Longstaff–Schwartz method:**

- Strengths: Flexible framework, can handle high-dimensional problems and complex payoff structures.
- Weaknesses: Choice of basis functions can significantly affect accuracy, may require a large number of simulations for accurate results, especially for options with long maturities or high volatility.

- **Tsitsiklis–Van Roy method:**

- Strengths: Can be more efficient than the Longstaff–Schwartz method in certain cases, provides a direct approximation of the option's value.
- Weaknesses: Similar to the Longstaff–Schwartz method, the choice of basis functions can significantly affect accuracy, may require a large number of simulations for accurate results, particularly for options with complex payoff structures or long maturities, **exponential blowup** in the number of basis functions needed for accurate approximation as the dimensionality of the problem increases, which can lead to computational challenges.

Scope and Claims

Conclusion

References

Longstaff, F. A., & Schwartz, E. S. (2001). Valuing American options by simulation: A simple least-squares approach. *The Review of Financial Studies*, 14(1), 113–147.

Tsitsiklis, J. N., & Van Roy, B. (2001). Regression methods for pricing complex American-style options. *IEEE Transactions on Neural Networks*, 12(4), 694–703.